Optimization of Electrode Configuration for Bladder-Targeted Temporal Interference Stimulation Using a Three-Dimensional Rat Model

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Temporal interference stimulation (TIS) is a noninvasive neuromodulation technique that uses two high-frequency currents with slightly different frequencies to generate a low-frequency modulation envelope within biological tissue. Effective targeting of deep organs using TIS depends strongly on electrode configuration and the resulting electric-field distribution. This study aimed to optimize electrode configuration for bladder-targeted TIS using a three-dimensional (3D) rat model.

A 3D anatomical rat model (104 × 44 × 40 mm) was implemented in COMSOL Multiphysics to evaluate electric-field distributions under different surface electrode configurations. Each tissue was assigned its corresponding electrical conductivity to represent tissue-specific electrical properties. Two stimulation frequencies of 8,000 and 8,010 Hz were applied at 3 mA. Five electrode configurations were compared using the electric-field magnitude within the bladder and surrounding tissues. Target selectivity was evaluated based on the ratio of the bladder electric field to off-target tissue fields. Electrode geometries, including circular, square, and rectangular electrodes, were additionally compared.

Electric-field distribution and bladder selectivity varied substantially among electrode configurations. Among the five configurations, Case 4 demonstrated the highest target selectivity. In this configuration, one electrode pair was positioned between the upper surface above the bladder and the lateral body surface, while the second pair was positioned between the lower surface below the bladder and the opposite lateral side. Case 4 produced a mean bladder electric field of 63.30 V/m and a surrounding-tissue field of 7.95 V/m, corresponding to a bladder-to-tissue selectivity ratio of 7.96. Among the evaluated electrode geometries, the 3 × 3 mm square electrode showed the highest selectivity (6.830), although only small differences were observed among the three geometries.

These results demonstrate that electrode configuration substantially influences bladder-targeted electric-field selectivity during TIS and support three-dimensional anatomical modeling as an approach for selecting electrode placement and geometry for targeted bladder neuromodulation.

Keywords : Temporal Interference Stimulation; Bladder Neuromodulation; Electrode Configuration; Three-Dimensional Rat Model; Electric Field; Selectivity